Jonathan Liu

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EDUCATION

University of Washington **Seattle, WA**
Bachelor of Science in Electrical and Computer Engineering | GPA: 3.93 *Sep 2024 – June 2027 (expected)*
Master of Science in Electrical and Computer Engineering (Combined BS/MS, Analog and Mixed-Signal) *Sep 2027 – June 2028*

- **Honors:** Frey Endowed EE Scholarship (2026–27), IEEE HKN Honor Society (Top 8% ECE), Tau Beta Pi Engineering Honor Society (Top 12.5%), Annual Dean's List (2024–2026).
- **Relevant Coursework: Analog IC Design (EE 435)**, **VLSI I (EE 476)**, Computer Architecture (EE 469), all in progress.

ANALOG IC DESIGN

Analog IC Tapeout, UW Analog Tapeout Sequence (EE 435/436/437) *Sep 2026 – Summer 2027*

- Designing an **analog IC for industrial-foundry fabrication** in a 3-quarter sequence sponsored by **Apple, AMD, and TSMC**.

5T OTA, Optical-Receiver TIA, and Inverter Layout in 45nm CMOS (EE 332) *Spring 2026*

- Designed a single-stage **5T OTA** in **Cadence Virtuoso** on **GPDK045** (45nm), hand-sizing the input pair, current-mirror load, and tail source from **gm/I_D** analysis.
- Diagnosed a first-pass sizing that left devices out of saturation; re-derived gm/I_D and optimized in **ADE Assembler** to **25.5 dB** gain, **1.20 GHz** GBW, and **1 V_{pp}** swing on **5.04 mW** across AC, DC, and transient sweeps.
- Closed the OTA into a shunt-shunt **transimpedance amplifier** for a 2 Gb/s optical receiver: **78 dBΩ** gain, **947 MHz** bandwidth (6x a direct-RC front end), opening a PRBS eye the RC design closed.
- Laid out a **CMOS inverter** in **FreePDK45** and passed **Calibre DRC** (0 violations on 167 checks) and **LVS**.

Semiconductor Engineering Practicum **July – Aug 2026**
Hiroshima University *Hiroshima, Japan*

- Selected for a U.S.–Japan semiconductor consortium with **Micron Technology** and **Tokyo Electron**; hand-drew a **CMOS inverter layout** in **Siemens L-Edit**, fabricated it in a cleanroom, and measured about **3.4 mA** nMOS saturation current on a **Keysight B1500A**.

RELEVANT EXPERIENCE

Hardware Engineering Intern **June – Sep 2026**
Trellis Robotics *Mountain View, CA*

- Designed a **4-layer, 11-IC motor-control and lithium battery-management PCB** from schematic to fabrication; diagnosed a failure that destroyed two prototypes, re-architected the battery protection, and brought the redesign up **the day it arrived with zero rework**.
- Architected a **next-generation Power-over-Ethernet robot PCB** that runs two motors and an industrial IMU from one cable within a **25.5 W budget**, and caught an ESP32 clocking erratum corrupting 2% of network frames **before fabrication**.
- Developed a **novel GPS-free 3D localization system** that maps a robot's path through metal pipe with **no GPS, compass, or fixed sensor mount**, using only a free-floating motion sensor and tether length; achieved **4.3 cm RMS accuracy** over 19 runs.
- Built the robot's **entire control stack from scratch**: **20,000+ lines of ESP32-P4 C firmware** (motor, battery, and sensor drivers; **1080p video at 27 fps**), an Ethernet-to-Wi-Fi bridge, and a browser dashboard; also built a **company-wide AI engineering assistant** that onboarded the entire company onto the stack.

Hardware Lead **Oct 2024 – Present**
Husky Robotics Drone Subteam *Seattle, WA*

- Led a 20+ person EE team and managed three custom PCBs in parallel: a **flight controller**, an electronic speed controller, and a power distribution board, including design reviews and the power and signal interfaces between them.
- Designed a 4-layer **STM32H743** flight controller in **Altium**: schematic, layout, 5 V/3.3 V regulation, current sensing, and the connector pinout for motors, radios, and sensors.
- Led **system integration** for a University Rover Challenge quadrotor: 6S power, 50 A ESCs, Pixhawk 6C, Jetson Orin Nano, **MAVLink**.

PROJECTS

- **AC-DC Boost Converter (June 2026):** designed, simulated in **LTspice**, and **breadboarded** a two-stage converter delivering an adjustable **+10–21 VDC** rail at **0.4%** load regulation under **100 mV** ripple.
- **Three-Band Active Equalizer (March 2026):** built an **LM348** op-amp equalizer; hand-derived transfer functions, validated in **LTspice**.
- **FPGARhythm (March 2026):** developed a hardware rhythm game in **SystemVerilog** on an Intel **Cyclone V** FPGA with **VGA** output.
- **Line-Following Car (Feb 2026):** built a **PID**-tuned Arduino car; **all-time fastest record** in the UW competition.

SKILLS

IC Design & Test: Cadence Virtuoso (schematic, layout, PEX), Calibre DRC/LVS, Spectre, ADE Explorer/Assembler, GPDK045/FreePDK45, Siemens L-Edit, LTspice/SPICE, gm/I_D sizing, cleanroom CMOS fabrication, B1500A I-V wafer probing.

PCB & Embedded: Altium Designer, KiCad, multi-layer layout, multi-rail PDN, power sequencing, IPC-2221/DFM, STM32 (CubeIDE/HAL), ESP-IDF, Arduino, UART/I^2C/SPI/CAN, battery management, motor drivers, buck converters, LDOs.

Digital & Software: SystemVerilog, FPGA (Quartus, ModelSim), C/C++, Python (NumPy/SciPy), MATLAB, Linux/Bash, Git.

Leadership: President, IEEE UW Student Chapter; **Events Officer**, Hong Kong Student Association at UW.

Languages: English, Mandarin, Cantonese, Japanese.